

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	AIDS	Date:	7-08-2026

Code of Conduct

I Viveka Sivanandham (192424128) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Viveka Sivanandham

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus: *"Data science is powerful, data science drives innovation, data science is evolving."*

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences. The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} \mid \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.
 2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
 3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
 4. Calculate the Entropy of the distribution $P(\text{is})=0.66$ and $P(\text{drives})=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.
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CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
 2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
 3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
 4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.
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CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
 - economic (120 occurrences)
 - growth (450 occurrences)
 - increases (210 occurrences)
 - employment (380 occurrences)

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.
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Answers

CASE STUDY 1: Smart Mobile Keyboard Prediction System

1. Compute the MLE of the Bigram Probability

The Maximum Likelihood Estimate (MLE) of a bigram is calculated using the formula:

From the given data:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$

So,

Answer: 1.0

Interpretation:

This means that every time the word "**data**" appears in the training corpus, it is followed by "**science**." Therefore, when a user types "**data**", the mobile keyboard can confidently predict "**science**" as the next word, improving typing speed and prediction accuracy.

2. Apply the Backoff Model

The sequence "**data science improves**" is not present in the corpus. Therefore, the trigram probability cannot be calculated.

Using the Backoff Model, the system first checks the trigram. Since it is unavailable, it backs off to the bigram "science improves", which is also not found. Finally, it backs off to the unigram "improves". As the word does not occur in the corpus, its probability is considered **0** (without smoothing).

Estimated Probability = 0

Why is Backoff important?

The Backoff Model helps the keyboard handle new or unseen word combinations instead of completely failing to make a prediction. In real-world applications, smoothing techniques assign a very small probability to unseen words, allowing the keyboard to continue providing useful suggestions.

3. Deleted Interpolation

Deleted Interpolation combines trigram, bigram, and unigram probabilities using weighted averages.

The given weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The trigram probability is:

The bigram probability is also:

The unigram probability of "is" is:

Applying the interpolation formula:

Final Probability = 0.57

Discussion:

Deleted Interpolation improves prediction by combining higher-order and lower-order language models. If the trigram information is unreliable, the bigram and unigram probabilities help maintain prediction accuracy, making the model more effective for sparse datasets.

4. Entropy Calculation

Entropy measures the uncertainty of predictions.

The formula is:

Given:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Substituting the values,

Answer: Entropy ≈ 0.92 bits

Interpretation:

The entropy value is less than 1 bit, indicating relatively low uncertainty. Since **"is"** has a much higher probability than **"drives,"** the keyboard is more confident in predicting **"is"** after the word **"science."**

CASE STUDY 2: AI-Powered Customer Support Chatbot

3. Assign Penn Treebank POS Tags

Sentence 1

Book a flight ticket now.

Sentence 2

This book is interesting.

Word	POS Tag
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This	DT
------	----

book	NN
------	----

is	VBZ
----	-----

interesting	JJ
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Justification:

In the first sentence, **"Book"** is used as a command, so it is tagged as a **Verb (VB)**. In the second sentence, **"book"** refers to an object, so it is tagged as a **Noun (NN)**. The surrounding words clearly indicate its grammatical role.

4. HMM Probability

The probability of **"book"** being a verb is calculated as:

Given,

- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{book} \mid \text{VB}) = 0.6$

Therefore,

Similarly,

Since **0.30 is greater than 0.20**, the HMM assigns **VB** to the word **"book."**

Answer: 0.30

Explanation:

The sentence starts with an instruction, making it more likely that **"Book"** is functioning as a verb. The probabilistic model correctly chooses the tag with the higher likelihood.

5. Rule-Based Tagging vs. HMM Tagging

Rule-Based Tagging depends on predefined grammar rules to assign tags, whereas HMM Tagging uses probabilities learned from training data. Rule-based systems are simple and easy to understand but often fail when words have multiple meanings. HMM Tagging considers context and is better at resolving ambiguity.

For a large-scale airline chatbot, **HMM Tagging** is the better choice because it can process a wide variety of user inputs and produce more accurate results in different contexts.

6. Importance of Standardized POS Tagsets

Standardized POS tagsets provide a consistent way of labeling words according to their grammatical roles. This helps the chatbot understand sentence structure, identify user intent, and generate more accurate responses. It also improves the performance of language models and ensures consistency across different NLP applications.

CASE STUDY 3: News Analytics and POS Tag Correction System

1. Apply the Transformation Rule

The initial POS tags are:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

According to the transformation rule:

Change NNS to VBZ if the previous word is tagged as NN.

Since growth is tagged as NN, the tag for increases changes from NNS to VBZ.

The corrected sequence is:

- economic/JJ
- growth/NN
- increases/VBZ

- employment/NN

This correction is appropriate because **"growth"** is the subject and **"increases"** is the verb describing the action.

2. Analyze the Initial Tagging Error

The initial tagger incorrectly labeled **"increases"** as a plural noun (**NNS**). However, in the sentence, **"increases"** functions as a third-person singular verb. After applying the transformation rule, the sentence becomes grammatically correct and accurately represents the intended meaning.

3. Compute the Word Frequency Distribution

Total word occurrences:

The frequency distribution is:

Word	Frequency	Probability
economic	120	$120/1160 = 0.103$
growth	450	$450/1160 = 0.388$
increases	210	$210/1160 = 0.181$
employment	380	$380/1160 = 0.328$

Explanation:

Words that occur more frequently provide stronger statistical evidence during tagging. Therefore, frequency information helps probabilistic taggers make better decisions and improves overall tagging accuracy.

4. Transformation-Based Tagging and Entropy

Transformation-Based Tagging improves the quality of POS tagging by applying learned correction rules after the initial tagging stage. In this example, it correctly changes **"increases"** from a noun to a verb based on the surrounding context. This results in a more grammatically correct sentence and higher tagging accuracy.

Entropy is used to measure the uncertainty of tag assignments. Before applying the transformation rule, the uncertainty is higher because the word **"increases"** could have multiple possible tags. After the rule is applied, the correct tag is confidently assigned, reducing entropy and improving the reliability of the tagging system.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____